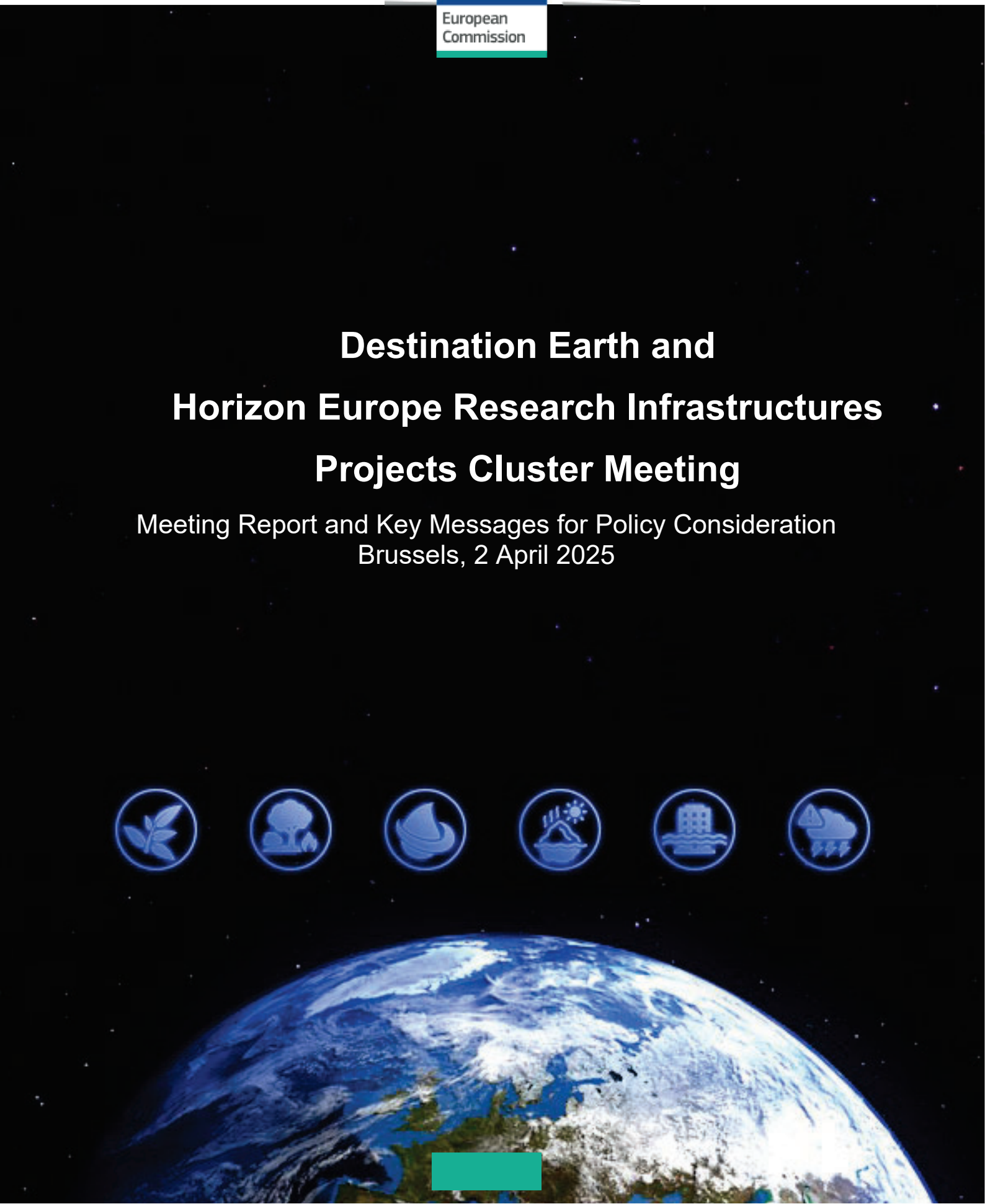
Destination Earth and Horizon Europe Research Infrastructures Projects Cluster Meeting

Meeting Report and Key Messages for Policy Consideration
Brussels, 2 April 2025



Written by Elena GARBARINO, Gustavo NAUMANN, Anna SANTORO, Maria VASCHIU
April – 2025

EUROPEAN COMMISSION

European Research Executive Agency (REA)
Unit C4 — Reforming European R&I and Research Infrastructures

E-mail: rea-he-infra@*ec.europa.eu*

European Commission
B-1049 Brussel

Destination Earth and Horizon Europe Research Infrastructures

Projects Cluster Meeting

Meeting Report and Key Messages for Policy Consideration

Brussels, 2 April 2025

European Research Executive Agency (REA)
Research Infrastructures

2025

Manuscript completed in July 2025

LEGAL NOTICE

This document has been prepared for the European Commission however it reflects the views only of the authors, and the European Commission is not liable for any consequence stemming from the reuse of this publication. More information on the European Union is available on the Internet (http://www.europa.eu).

PDF ISBN 978-92-95234-69-7 doi: 10.2848/7804382 JW-01-25-024-EN-N

Luxembourg: Publications Office of the European Union, 2025

© European Union, 2025



The reuse policy of European Commission documents is implemented by the Commission Decision 2011/833/EU of 12 December 2011 on the reuse of Commission documents (OJ L 330, 14.12.2011, p. 39). Except otherwise noted, the reuse of this document is authorised under a Creative Commons Attribution 4.0 International (CC-BY 4.0) licence (https://creativecommons.org/licenses/by/4.0/). This means that reuse is allowed provided appropriate credit is given and any changes are indicated.

For any use or reproduction of elements that are not owned by the European Union, permission may need to be sought directly from the respective rightholders.

Table of Contents

Summary 6
Destination Earth (DestinE) Initiative and Horizon Europe Funding Programme 8
Horizon Europe Funded Projects 8
InterTwin Project 10
BioDT Project 10
DTGeo Project 11
TerraDT Project 12
UrbAnair Project 12
WeatherGenerator Project 13
Other relevant projects funded under Horizon 2020 and Horizon Europe 13
NextGems Project 14
EERIE Project 14
EDITO: European Digital Twin Ocean 15
EDITO-Infra 16
EDITO-Model Lab 16
EU Contribution 16
Overview of the event 18
Breakout Session 1: Collaboration and Interaction for DestinE 18
Breakout Session 2: Challenges and Opportunities for DestinE 19
Conclusions 20

Summary

In recent years, the increasing frequency and intensity of extreme weather events, such as floods, droughts, hurricanes, and heat waves, have underscored the urgent challenges posed by climate change. These events present significant risks to human lives, ecosystems and the global economy. As a result, there is an urgent need for comprehensive research and immediate action to develop strategies and technologies that can protect vulnerable communities and ecosystems from the impacts of global warming.

In response to these pressing challenges, Destination Earth (DestinE) is an ambitious initiative launched by the European Commission aimed at developing a highly accurate digital model of the Earth: a digital twin of the Earth. This digital twin model monitors and simulates natural phenomena, hazards and the related human activities, enabling policymakers and researchers to better understand environmental challenges and anticipate climate change impacts. With the advent of big data analytics and supercomputing, DestinE will help key actors gain insights into how to integrate artificial intelligence (AI) into environmental research and policymaking.

Climate change remains a high priority on the European Commission's agenda and DestinE is an essential pillar of the European Commission's efforts towards the Green Deal and Digital Strategy. This is reflected in funding opportunities in EC programmes, notably in the Horizon Europe Research Infrastructures work programme (INFRA), that funded six projects between 2021 and 2024.

In this context, the European Research Executive Agency (REA) and the Directorate General for Communications Networks, Content and Technology of the EC (DG CNECT) jointly organised a thematic meeting involving the six funded INFRA projects contributing to DestinE and representatives of the DestinE implementing entities, European Centre for Medium-Range Weather Forecasts (ECMWF), the European Space Agency (ESA) and the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT).To explore synergies with other parts of Horizon Europe, four more projects funded between 2020 and 2024 were invited to the event (see Figure 1).

The event achieved two main objectives.

Firstly, it created a venue for project participants to meet and discuss their contributions to the twin transition, addressing both scientific and technical challenges, exploring synergies with other EC-funded projects, anticipating future challenges, and examining the integration of AI.

Secondly, it brought together project representatives with policy makers and DestinE's implementing entities to further address funding needs in the context of the ongoing discussion taking place at European level on the next Multiannual Financial Framework (MFF) and research Framework Programme (FP10).

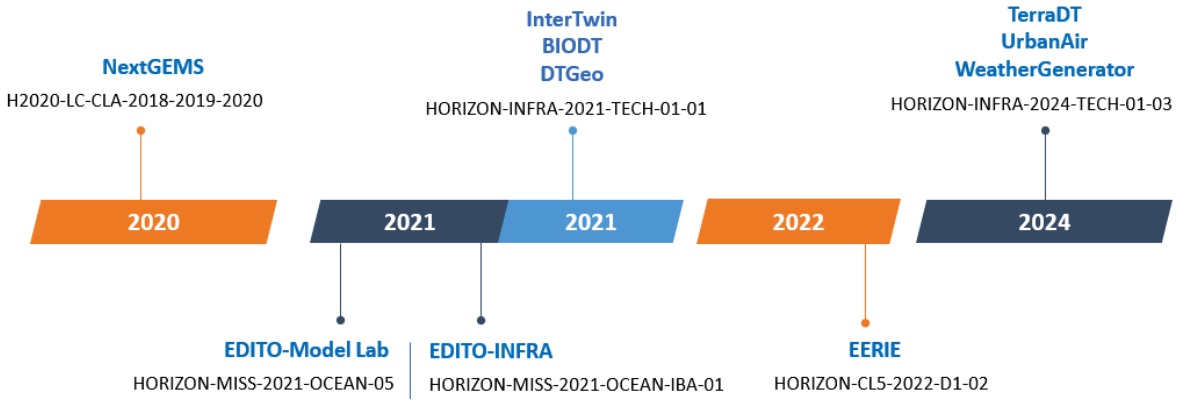


Figure 1

Two aspects of DestinE implementation were discussed: collaboration and interaction, and challenges and opportunities.

The **key messages** from the event are presented below.

- **Collaboration and interaction among projects and organisations working on DestinE are crucial**; e.g. sharing code and data, organising joint work strands and activities, and facilitating ground-level technical discussions;
- **Sustainability and long-term planning are essential**; providing access and funding to support project continuation, and creating long-term mechanisms for sharing data and code on a voluntary basis;
- **International connections and national projects are vital**; i.e. stronger connections with countries/areas like Canada, the Caribbean, Africa, and the US are needed;
- **Challenges and opportunities of climate models**. Challenges: integration of data with coupled models, the need for accredited views of models, and security concerns, including the need for trustworthy AI, validation, and testing. Opportunities: development of domain-focused High Performance Computing centres, the creation of a public infrastructure to tap innovative potential;
- **Brokering science to policy making is critical;** making science accessible to policymakers at all levels is essential, using interactive AI tools, improving the quality of upstream data, and making information accessible to general models and physical models.

Destination Earth (DestinE) Initiative and Horizon Europe Funding Programme

DestinE is a flagship initiative of the EU arising from the urgency of acting in the context of the increasingly frequent and severe weather events. For example, the record heatwave of 2022 has been linked to between 60 000 and 70 000 premature deaths in Europe. If decisive action is not taken now, economic losses from coastal floods alone could exceed EUR 1 trillion by the end of this century[1].

DestinE aims to gradually develop a highly accurate digital twin of the Earth. It will revolutionise the modelling, simulation and monitoring of natural phenomena, hazards and related human activities, supporting Europe's role as a global leader in fighting and mitigating climate change. DestinE will unlock unprecedent digital capabilities for adapting to climate change, forecasting extreme events and interactive use of high-resolution climate data. DestinE builds on EU's investment in high-performance computing, the wealth of available data including massive Earth observation data sources, and on the European excellence in data-driven innovation and breakthrough AI technologies.

DestinE will provide dedicated "what-if" scenarios to decision-makers, in support of climate change adaptation policies and mitigation scenario testing at decadal timescales, aiming at a real breakthrough in terms of reliability at EU, national, regional and local levels.

The European Commission leads the implementation of DestinE in coordination with Member States and Associated Countries, partnering with major European intergovernmental organisations with primary expertise, i.e.:

- European Space Agency (ESA)
- European Centre for Medium-Range Weather Forecasts (ECMWF)
- European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT)

Horizon Europe Funded Projects

Horizon Europe is the EU's key funding programme for research and innovation with a budget of EUR 95.5 billion running from 2021 to 2027. Horizon Europe consists of three main pillars and one horizontal activity. Research Infrastructures are funded under Pillar I, which is dedicated to Excellent Science.

Within this framework, the European Commission's Research Infrastructures Work Programme plays a pivotal role in advancing the European Research Area by supporting the development and sustainability of state-of-the-art research facilities across the continent. This programme is integral to Horizon Europe's mission to foster excellence in research and innovation by providing access to world-class infrastructures and promoting their integration

[1] European Climate Risk Assessment - Executive summary

across various scientific disciplines. The programme aims to facilitate cross-border collaboration and drive scientific breakthroughs that address key societal challenges by enhancing transnational access, developing robust e-infrastructures, and strengthening data management capabilities.

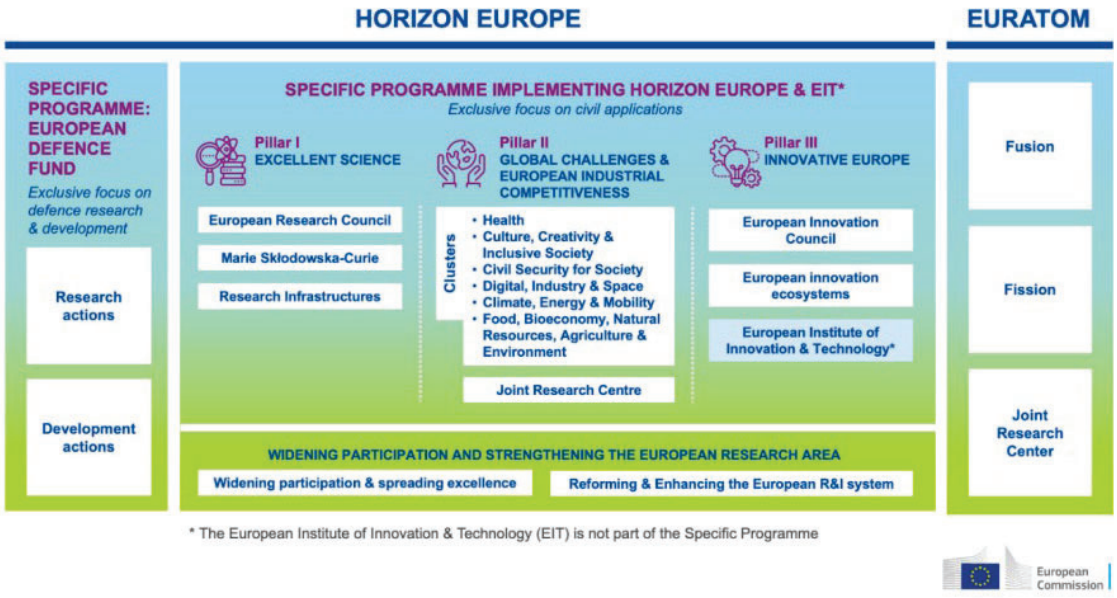


Figure 2

Under Horizon Europe Research Infrastructures programme, two pivotal calls have been issued to date, targeting the central themes of DestinE.

HORIZON-INFRA-2021-TECH-01-01 - Interdisciplinary digital twins for modelling and simulating complex phenomena at the service of research infrastructure communities

The call for proposals HORIZON-INFRA-2021-TECH-01-01 launched in 2021, aimed at developing digital twins providing advanced modelling, simulation and prediction capabilities to RIs and their research communities through a convergent use of advanced digital technologies such as high-performance computing, software, AI methods and data analytics. The work under this topic aimed at reaching a sufficiently high TRL level (6-7) to be considered for integration into operational activities of for example existing research infrastructures, the EOSC (European Open Science Cloud) platform, and undertaken in related fields.

The funded projects were InterTwin, BioDT and DTGeo.

InterTwin Project (Website)

InterTwin co-designs and implements the prototype of an interdisciplinary Digital Twin Engine (DTE) – an open-source platform based on open standards that offers the capability to integrate with application-specific Digital Twins (DTs). Its functional specifications and implementation are based on a co-designed interoperability framework and conceptual model of a DT for research – the DTE blueprint architecture. The ambition of InterTwin is to create consensus on a common approach to the implementation of DTs that is applicable across the whole spectrum of scientific disciplines and will facilitate developments and interoperability across different DTs.

- Interdisciplinary Digital Twin Engine (DTE) federated infrastructure integrated with EOSC and EU Data Spaces.
- Interoperability framework including guidelines, specifications, and blueprint architecture.
- Toolkit for AI workflow and method lifecycle management.

BioDT Project (Website)

The Biodiversity Digital Twin (BioDT) project offers advanced simulation and prediction models to tackle critical global biodiversity challenges. The project is designed to address complex biodiversity dynamics through practical use cases, providing valuable insights and tools for conservation and restoration efforts.

BioDT harnesses the immense computational power of the LUMI Supercomputer, one of the fastest and most powerful computing systems in the world. By integrating FAIR (Findable, Accessible, Interoperable and Reusable) data principles with state-of-the-art digital infrastructure, predictive modelling, and AI solutions, BioDT facilitates the development of precise and reliable models. These models are crucial for creating evidence-based strategies aimed at preserving and restoring biodiversity on a global scale.

- Create simulation and prediction models for global complex biodiversity challenges.
- Prototyping Digital Twins on practical use cases for conservation and restoration efforts.
- Four main groups of biodiversity challenges tested: species response to environmental change, and interactions with each other and with humans, genetically detected biodiversity, dynamics and threats from and for species of policy concern.

DTGeo Project (Website)

The digital twin for GEOphysical extremes (DT-GEO) aims to analyse and forecast the impact of tsunamis, earthquakes, volcanoes, and anthropogenic seismicity. The project builds upon existing knowledge from other European projects and Centres of Excellence and pushing it forward.

To achieve its goals, DTGeo will build the components of a digital twin- virtual replica of physical systems that combine real-time data streams and high-fidelity models. Digital twins are built on interactive digital platforms, which allow for continuous observation and simulation of supercomputing tasks to recreate present and future scenarios. It capitalises on the development of supercomputing and technical capabilities to carry out advanced, real-time modelling and analysis for diagnostics and forecasting.



- Design and create a pre-operational prototype of a DT on geophysical extremes for its future integration into the DestinE initiative and test it in operational environments at 13 demonstrator sites of particular relevance located in Europe and beyond.
- Apply 12 DTs components addressing phenomena to conduct precise data-informed early warning systems, forecasts, and hazard assessments across multiple time scales.
- Supply a flexible framework for EOSC-enabling and FAIR-validation of project assets and outcomes and its integration in the European Plate Observing System and HPC/virtual cloud computing Research Infrastructures.

HORIZON-INFRA-2024-TECH-01-03 - New digital twins for Destination Earth

The second DestinE - RI call for proposals was launched in 2024 and switched the focus from transition to integration. The expected EU contribution per project was of EUR 15 million for the beneficiaries to be able to achieve TRL 6 or higher by the end of the project. Having already a firm foundation to build on from the first call, project results were expected to contribute to all the following aspects:

- Emergence of new science-based digital twin infrastructures to be gradually integrated in DestinE;
- Establishment of new digital standards for software (including simulation and simulation-observation data fusion) and data for DestinE;
- Standards- and science-based approach for modelling, predicting and assessing the Earth systems and their socio-economic impact.

The funded projects were TerraDT, Urban Air and WeatherGenerator.

TerraDT Project (Website)

TerraDT is focused on developing a high-resolution Digital Twin of Earth's cryosphere (frozen regions), land surface, and aerosols. The project addresses key challenges in climate modelling by improving the representation of these critical Earth system components. As part of the DestinE initiative, TerraDT integrates modular, interoperable components to enhance climate simulations at a global scale. These advancements aim to improve the accuracy and reliability of climate impact assessments, supporting informed decision-making for climate adaptation and mitigation.



- Integrate expertise in Earth system modeling, supercomputing and impact assessments from all over Europe.
- Develop an interactive user interface, through which users can ask concrete what-if questions.
- Provide very high resolution of 10 km climate projections, using computing platforms such as LUMI supercomputer and MareNostrum 5.

UrbAnair Project (Website)

UrbanAIR project aims at developing a new urban digital twin. This will support decision makers in urban areas to deal with urban design dilemmas in atmospheric heat and air quality to maximise the health, and socio-economic wellbeing and economic prosperity of its citizens affected by climate change. UrbanAIR will provide critical tools for climate adaptation and hazard control through urban design and planning including very high-resolution model components of the complex atmospheric Earth system.



- Yielding a dynamic, user-friendly infrastructure integrated into the DestinE infrastructure that empowers municipalities and industries to face urgent urban climate risks.
- The scales in the atmospheric models cover the full range from the regional to the neighbourhood level.
- The multiscale approach is achieved through the development of software interfaces for the modular coupling of atmospheric models.

WeatherGenerator Project (Website)

This project will build the WeatherGenerator, the Foundation Model of the Earth system that will serve as a new digital twin for DestinE. The WeatherGenerator will be based on representation learning and create a general and versatile tool that models the dynamics of the Earth system based on a large variety of Earth system data. It will be task-independent and will improve results for a wide range of machine learning applications when compared to task specific machine learning tools. It will also be more resilient for climate applications when the underlying data distributions are changing, and it will lead to a significant reduction in computational costs and faster turnaround times.

Weather Generator

- Develop a machine-learned digital twin of the Earth system, to serve as a foundation model for weather, climate, and Earth system science applications.
- Deepen Earth system understanding.
- Focus on enhancing sectorial impacts such as renewable energy forecasting (wind/solar/hydropower), public safety through early warning systems (floods/heatwaves), health risk mitigation (heat stress/disease modelling), agricultural resilience (crop failure prediction), and water resource management (flood prediction/river inflow modelling).

Other relevant projects funded under Horizon 2020 and Horizon Europe

Other projects funded under HE Pillar II that are contributing significantly to the DestinE objectives are considered for synergies with the projects funded under the two Research Infrastructures calls:

- NextGEMS (Next Generation Earth Modelling Systems), funded under Horizon 2020, contributes to climate action
- EERIE (European Eddy RIch Earth System Models), funded under Horizon Europe Cluster 5 within the 2022 call, contributes to climate science and solutions
- The two sister projects EDITO - (1) EDITO-Infra and (2) EDITO-Model Lab, funded under Horizon Europe, contributes to the Mission Restore our Ocean and Waters.

EU contribution to the three projects amounts to EUR 28 786 575.73.

NextGems Project (Website)

NextGEMS is a collaborative European project that taps expertise from fourteen European Nations to develop two next generation (storm-resolving) Earth-system Models. Through breakthroughs in simulation realism, these models will allow us to understand and reliably quantify how the climate will change on a global and regional scale, and how the weather, including its extreme events, will look like in the future.

Storm-resolving models differ from traditional climate models by operating at much higher resolutions, by using a fifty-fold finer horizontal grid, allowing them to explicitly represent the circulations that make the storms in the atmosphere, the eddies in the ocean, and cracks in the ice. These models are important to reduce systematic and long-standing errors in conventional climate models, providing enhanced predictions of extreme weather events.

- Create two storm-resolving Earth System Models (SR-ESMs) capable of representing the coupled climate system more accurately, performing 30-year climate projections, and integrating Earth-system processes like the carbon cycle and aerosols.
- Use SR-ESMs to test emerging and long-standing hypotheses underpinning current understanding of climate change.
- Build integrated ESM user communities by involving a wider scientific and non-scientific community in model development and promoting knowledge co-production.

EERIE Project (Website)

The project, running from January 2023 to December 2026, will reveal and quantify the role of ocean mesoscale processes in shaping the climate trajectory over seasonal to centennial time scales. To this end EERIE (European Eddy-RIch ESMs) will develop a new generation ESMs that are capable of explicitly representing a crucially important, yet unexplored regime of the Earth system – the ocean mesoscale. Leveraging the latest advances in science and technology, EERIE will substantially improve the ability of such ESMs to faithfully represent the centennial-scale evolution of the global climate, especially its variability, extremes and how tipping points may unfold under the influence of the ocean mesoscale.



- Model and understand the climate impacts of ocean mesoscale processes, including ocean eddies.
- Develop novel experimental designs, aiming to enrich the Coupled Model Intercomparison Project (CMIP) in support of future national and international climate change assessments, hence sustain and enhance European cooperation and leadership in climate sciences.
- Halve the time and energy needed to produce climate information with the highest level of process fidelity.
- Increase the reliability of regional climate change attribution by developing improved estimates of background natural variability.

EDITO: European Digital Twin Ocean

A Digital Twin of the Ocean is a virtual representation of all marine and coastal environments around the globe. It gives access to real-time and historical observations from thousands of sensors across the world ocean, as well as numerous satellites in space.

The European Commission launched the European Digital Twin of the Ocean at the One Ocean Summit in Brest, France, in February 2022. As a main element of the Digital Ocean Knowledge System under the European Union's "Mission Restore our Ocean and Waters", its ambition is to make ocean information readily available to all – international policymakers, national governments, researchers, innovators, businesses, entrepreneurs, activists, and citizens.

The European Digital Twin of the Ocean will provide an innovative set of user-driven, interactive and decision-making tools, backed by the best science and data. Its core development is underway with the funding from the EU through the European Digital Twin Ocean (EDITO) R&D projects. These two sister projects, namely EDITO-Infra and EDITO-Model Lab, will build the operational infrastructure of EDITO. EDITO's further aim is to construct and evolve a thriving digital ecosystem through a number of other relevant, complementary actions, aiming for an operational Digital Twin of the Ocean.

EDITO-Infra (Website)	***EDITO-Model Lab (Website)***
A 2-year project that builds the public infrastructure backbone for EDITO by integrating key data service components (among which Copernicus Marine Service and EMODnet), and by sharing cloud processing capabilities and software into a single digital framework.	A 3-year project aiming to develop the next generation of ocean models, combining artificial intelligence and high-performance computing to be integrated into the EDITO public infrastructure, providing access to focus applications and simulations of different what-if scenarios for the sustainable management of marine environments.

EU Contribution

The EU contribution through HE to the projects mentioned above amounts to about EUR 107 million, with an increase in the amount of funding granted for the projects under INFRA TECH calls from 2021 (approximately EUR 11 million per project) to 2024 (approximately EUR 15 million per project), which reiterates the European Commission's commitment to support the efforts of research communities towards DestinE goals.

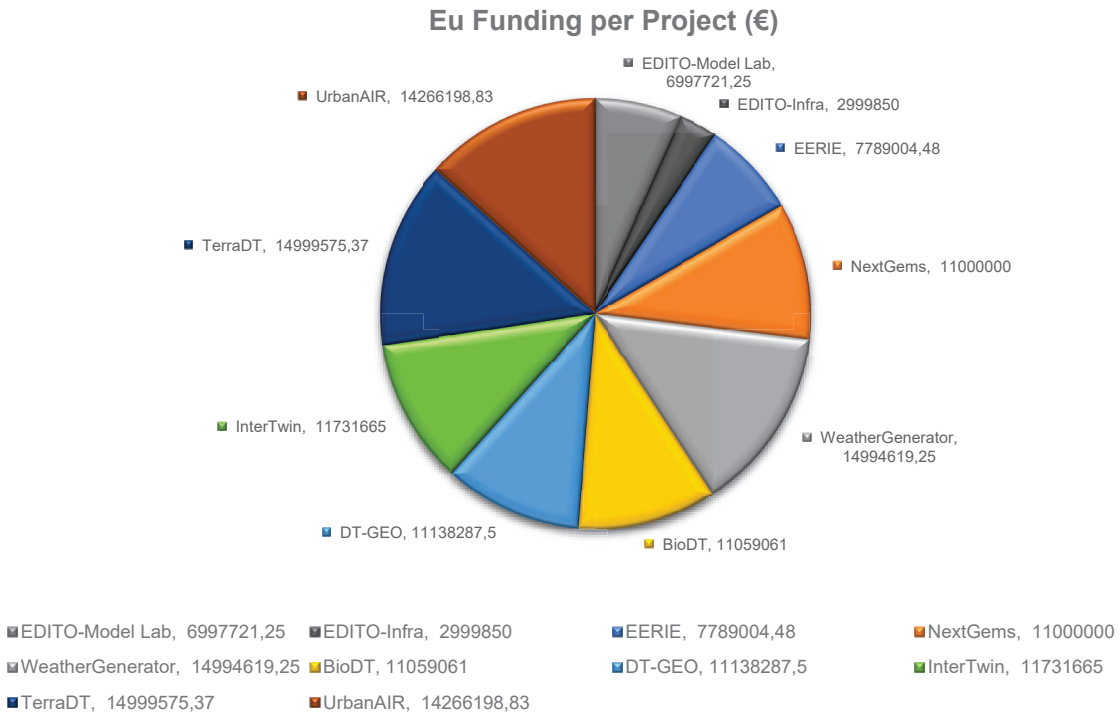


Figure 3

Some organisations and countries are taking leading roles in what concerns participation in the DestinE initiative. Germany, Italy and the UK lead in terms of participation, each having at least 7 participations in the selected projects, as well as in coordinating roles to which other countries have joined efforts (Netherlands, Finland, Spain).

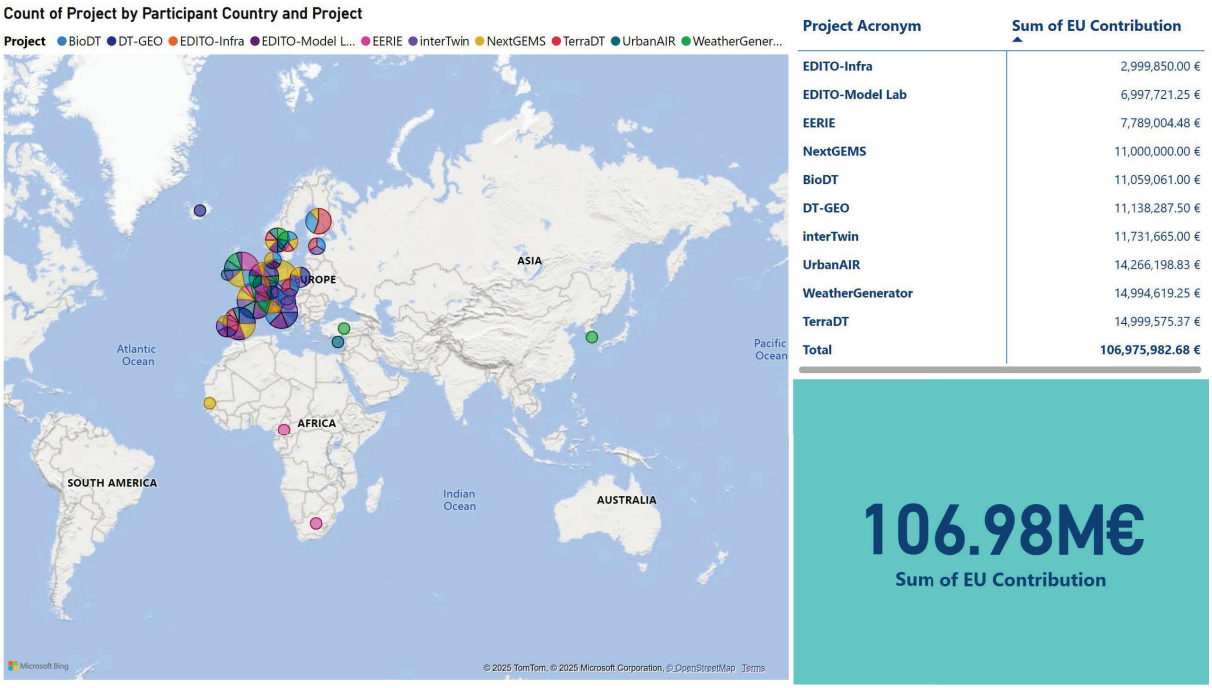


Figure 4

Barcelona Supercomputing Center (BSC CNS) and the Federal Institute of Technology Zurich (ETH Zurich) are leading the way with 6 participations, followed closely by Centre National de la Recherche Scientifique (CNRS) and the European Centre for Medium-Range Weather Forecasts (ECMWF) with 5 participations.

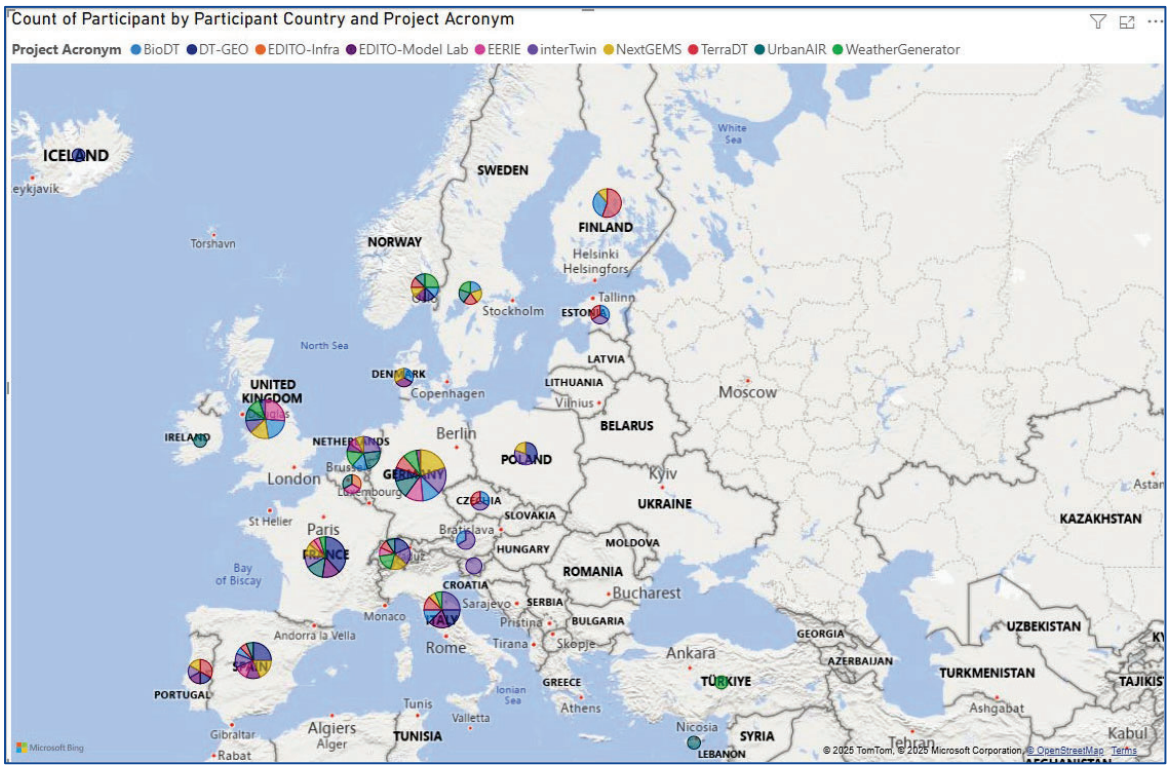


Figure 5

Overview of the event

The event, organised by REA and DG CNECT, convened experts from various sectors to exchange insights on the DestinE initiative. The event focused on assessing the current progress of DestinE implementation, encompassing supporting R&D projects, scientific and technical milestones, challenges, and fostering synergies among projects and other stakeholders. The event featured presentations by the implementing entities of DestinE, such as the European Centre for Medium-Range Weather Forecasts (ECMWF), the European Space Agency (ESA), and the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT), alongside contributions from representatives of selected Horizon Europe Research Infrastructure projects.

Further, two breakout sessions were held to facilitate in-depth discussions among experts. These sessions were structured around specific questions aimed at exploring collaboration, interaction, and sustainability within the DestinE initiative, as well as identifying key challenges and opportunities.

Participants had the opportunity to share their experiences and insights, contributing to a comprehensive dialogue on the initiative's future directions and prospects, and further contribute to the discussion around the new Multiannaula Financial Framework (MFF) and research Framework Programme (FP10).

Breakout Session 1: Collaboration and Interaction for DestinE

This session was guided by four key questions, which aimed to explore the current state of collaboration, interaction, and sustainability of the DestinE initiative.

1. Which other parts of the Horizon Europe or other EC funded programmes you have been collaborating with or intend to collaborate with? Which international connections are the most important? How about national projects and initiatives?

2. What is your main connection to the DestinE initiative and the three implementing entities (ECMWF, ESA, EUMETSAT) and/or their contractors? How do you interact? Can you share best practices?

3. What are the best ways to interact with other projects working on DestinE digital twins (joint work strands and activities, workshops, conference etc.)? Can you share best practices?

4. What is your project's contribution to the achievement of the objectives of the DestinE in the short term (project lifetime) and long term (after the project)? Where do you see the main challenges and how do you intend to sustain the projects results?

Box 1: Collaboration and Interaction for DestinE

Collaboration and International Connections

The participants highlighted the importance of collaboration with other Horizon Europe projects, such as sister projects on digital twins, EERIE, NextGEMS, and EOSC. International connections with countries like Canada, the Caribbean, Africa, and the US were also deemed crucial, with some experts mentioning the potential for standardisation and harmonisation of digital twins. National projects and initiatives, such as those in France, Italy, and Germany, were also discussed as potential areas for collaboration.

Connection to DestinE Initiative

The participants emphasised the need for clear connections to the DestinE initiative and its implementing entities, including ECMWF, ESA, and EUMETSAT. Some experts mentioned the importance of accessing and using DestinE data, as well as providing data to the initiative. Others highlighted the need for mechanisms to share codes and data among projects, as well as strategies for linking data projects and ensuring project sustainability.

Interaction with Other Projects

The participants discussed various ways to interact with other projects working on DestinE digital twins, including joint work strands and activities, workshops, and conferences. Some experts suggested organising interactive workshops and scientific conferences to facilitate technical discussions. Others mentioned the importance of cataloguing project outputs and promoting highly specialised interactions for technical details.

Sustainability and Long-Term Objectives

The participants emphasised the need for sustainability and long-term planning for the DestinE initiative. Some experts highlighted the importance of requesting data sharing plans in calls and providing access and funding to support project continuation. Others mentioned the need for mechanisms to share data and codes on a voluntary basis, as well as strategies for linking data projects and ensuring project sustainability. The experts also discussed the challenges of accessing resources, such as HPC access, and the need for open-source communities to continue support.

Breakout Session 2: Challenges and Opportunities for DestinE

This session was guided by four key questions, which aimed to explore the main challenges and opportunities for the DestinE initiative.

1. What are the main challenges you see in the DestinE initiative? How are you addressing those in your project?

2. How do you broker science to all levels of policy making? Make it as easy as possible for them to make related/informed decisions.

3. What are the low-hanging fruits in the DestinE initiative? How to integrate the data with coupled models?

4. How far can climate models go? What are the limitations and opportunities for improvement?

Box 2: Challenges and Opportunities for DestinE

Main Challenges and Opportunities

The participants highlighted several main challenges for the DestinE initiative, including the integration of data with coupled models, the need for accredited views of models, and the importance of considering security concerns. Some experts mentioned the challenge of getting advantage of public infrastructures with data and integrating full mitigation measures into adaptation ones. Others emphasised the need to discriminate between different end-users, notably expert users (such as researchers, meteorological offices) and non-expert users (such as citizens and policy makers), and to make use of the power of AI to support decision-making.

Brokering Science to Policy Making

The participants discussed the importance of brokering science to all levels of policy making, making it easy for policy makers to make informed decisions. Some experts suggested using interactive AI tools, such as the GPT tool used in Copernicus, to support policy makers in choosing scenarios and making decisions. Others emphasised the need to improve the quality of upstream data and to make information accessible to general models and physical models.

Immediate opportunities and Integration of Data

The participants identified several opportunities in the DestinE initiative, including the integration of data with coupled models and the development of domain-focused HPC centres. Some participants mentioned the importance of creating a public infrastructure that taps the innovative potential of people to address challenges. Others emphasised the need for concrete examples and proper problem statements to address specific challenges.

Limitations and Opportunities for Climate Models

The participants discussed the limitations and opportunities for climate models, including the need for trustworthy AI and the importance of validation and testing. Some experts mentioned the challenge of scaling climate models and the potential of using Large Language Models (LLMs) to bring different data together. Others emphasised the need to create a public infrastructure that allows people to "reason" with the data and to make breakthroughs with current powerful infrastructure.

Conclusions

In conclusion, this project cluster event highlighted the importance of collaboration, interaction, and sustainability for the DestinE initiative.

Experts in the field emphasised the importance of creating a public infrastructure that taps into the innovative potential of scientific results to address challenges and make breakthroughs.

Some challenges ahead include creating strong connections between the initiative's implementing entities, EU funded projects, national projects and initiatives and international players outside the EU.

Moreover, there is a need to establish a clear mechanism for sharing code and data among projects, for highly specialised technical discussions and cataloguing project outputs, including

the integration of data with coupled models, the need for accredited views of models, as well as the importance of considering security concerns.

A continuous dialogue between scientists and policymakers at all levels of policy-making was emphasised to enhance the quality of upstream data and ensure information is accessible for the development and improvement of all types of models and modeling communities.

Horizon Europe Cluster Event on Digital Twins for Destination Earth

2 April 2025 – 9:30-16:00

Covent Garden, Place Rogier 16, Brussels

AUDITORIUM (COVE A2 4-15)

Programme

08:45 - 09:30		*Registration and welcome coffee*
09:30 - 09:45	o	**Opening and welcome** Grazyna PIESIEWICZ, Head of Unit High Performance Computing and Applications, DG CNECT Sari VARTIAINEN-MATHIEU, Head of Unit Reforming European R&I and Research Infrastructures, REA
09:45 - 10:30	o	**Current state of play by DestinE** Irina SANDU, Director of Destination Earth, ECMWF Barbara BORGIA, DestinE System Integration and Validation Engineer, ESA Lothar WOLF, Head of Digital Solutions & SAF, EUMETSAT
10:30 - 11:00	o	**HORIZON-INFRA-2021-TECH-01-01** ***Interdisciplinary digital twins for modelling and simulating complex phenomena at the service of research infrastructure communities*** Andrea MANZI, STICHTING EGI, **InterTwin** Damien LECARPENTIER, CSC, **BioDT** Ramon CARBONEL, CSIC, **DTGeo**
11:00 - 11:15		*Coffee break*
11:15 - 11:45	o	**HORIZON-INFRA-2024-TECH-01-03** ***New digital twins for Destination Earth*** Jenni KONTKANEN, CSC, **TerraDT** Femke VOSSEPOEL, DELFT UNIVERSITY, **Urbanair** Peter DUEBEN, ECMWF, **WeatherGenerator**

11:45 – 12:15	**O**	**Presentations by other projects** Bjorn STEVENS, Max Planck Institute for Meteorology, **NextGems** Thomas JUNG, AWI, **EERIE** Alain ARNAUD, Mercator Ocean International, **EDITO**
12:15 – 13:30		*Networking Lunch*
13:30 – 14:45	**O**	**Breakout sessions** **1 – Best practices, collaborations and legacy** Moderated by: Andrea HUTTERER, Head of Sector RI, REA **2 – Future research priorities** Moderated by: Liina MUNARI, Deputy Head of Unit, DG CNECT
14:45 – 15:00		*Coffee Break*
15:00 – 15:45	**O**	**Round table discussion** Moderated by: Liina MUNARI, Deputy Head of Unit, DG CNECT
15:45 – 16:00	**O**	**Conclusions and next steps** Grazyna PIESIEWICZ, Head of Unit High Performance Computing and Applications, DG CNECT Sari VARTIAINEN-MATHIEU, Head of Unit Reforming European R&I and Research Infrastructures, REA



Project	Participants
ECMWF	Irina Sandu, Daniel Thiemert
ESA	Barbara Borgia, Franka Kunz
EUMETSAT	Lothar Wolf
InterTwin	KRAKOWIAN Małgorzata, Manzi Andrea
BioDT	LECARPENTIER Damien
DT-GEO	Stefan Wiemer, Ramon Carbonell
TerraDT	Jenni Kontkanen, Devaraju Narayanappa
UrbanAIR	Femke Vossepoel, Filip Lefebre, Natalie Theeuwes
WeatherGenerator	Dueben Peter, Christian Lessig
NextGems	Bjorn STEVENS
EERIE	Thomas JUNG
EDITO-Infra	Alain ARNAUD, Piero Lanucara
REA	VARTIAINEN-MATHIEU Sari, HUTTERER Andrea, GARBARINO Elena, NAUMANN Gustavo, SANTORO Anna, ANDREDAKIS Ioannis, SKONDRA Ourania, KAKARANTZAS Sotirios, FASANO Gelsomina
EC CNECT	MUNARI Liina, ADAMOVIC Marko, FALK Ulrike, TSITLAKIDIS Charalampos, PIESIEWICZ Grazyna, LE-SAUX Bertrand
DG RTD	CONSTANTIN Mihaela, IMMLER Franz, DRABICKA Katarzyna, RUIZ RAMOS Margarita, Federico Fierli
CINEA	TAVARES Richard, PALIALEXIS Andreas

GETTING IN TOUCH WITH THE EU

In person

All over the European Union there are hundreds of Europe Direct information centres. You can find the address of the centre nearest you at: https://europa.eu/european-union/contact_en

On the phone or by email

Europe Direct is a service that answers your questions about the European Union. You can contact this service:

– by freephone: 00 800 6 7 8 9 10 11 (certain operators may charge for these calls),

– at the following standard number: +32 22999696, or

– by email via: https://europa.eu/european-union/contact_en

FINDING INFORMATION ABOUT THE EU

Online

Information about the European Union in all the official languages of the EU is available on the Europa website at: https://europa.eu/european-union/index_en

EU publications

You can download or order free and priced EU publications from: https://op.europa.eu/en/publications. Multiple copies of free publications may be obtained by contacting Europe Direct or your local information centre (see https://europa.eu/european-union/contact_en).

EU law and related documents

For access to legal information from the EU, including all EU law since 1952 in all the official language versions, go to EUR-Lex at: http://eur-lex.europa.eu

Open data from the EU

The EU Open Data Portal (http://data.europa.eu/euodp/en) provides access to datasets from the EU. Data can be downloaded and reused for free, for both commercial and non-commercial purposes.

JW-01-25-024-EN-N



Publications Office
of the European Union

ISBN 978-92-95234-69-7